

Cumulative frequency tables and histograms



1

a 26

b 6

c 18

2

Score (x)	Frequency (f)	Cumulative frequency (cf)
18	10	10
19	9	19
20	3	22
21	7	29
Total:	29	

3

a

Number of Sightings	Number of Locations (f)	Cumulative frequency (cf)
7	4	4
8	3	7
9	5	12
10	2	14
11	5	19
12	1	20

b 6

c 14

d 9

4

a 2

b 12

c

Score (x)	Frequency (f)	Cumulative frequency (cf)
2	1	1
3	2	3
4	5	8
5	5	13
6	5	18
7	9	27
8	7	34
9	5	39
10	8	47
11	1	48
12	2	50



d

7

e 11

f 18

5

Score	Frequency	Cumulative frequency
25	10	10
26	16	26
27	14	40
28	20	60
29	16	76
30	10	86

6

a 14

b 2

c 0

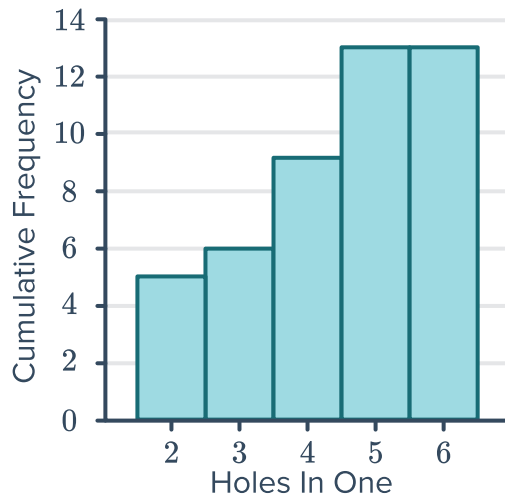
d 71.4%

7

a



b 45



c 6

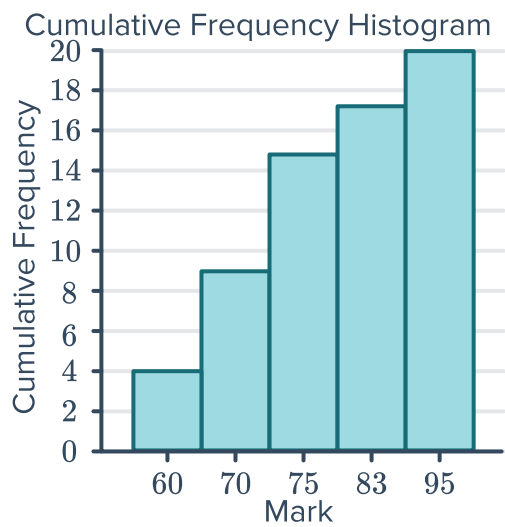
8

a The difference between the heights of two adjacent columns.

b 7

9

a



b 11

c 9

d 2

e 75

f 74.55

g 75

10



a

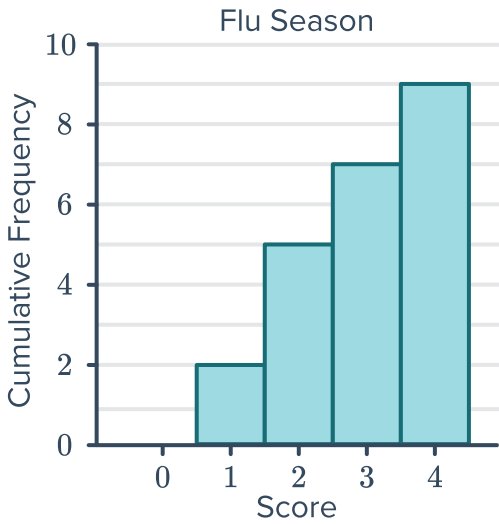
Score (x)	Frequency (f)	fx	Cumulative frequency
0	0	0	0
1	2	2	2
2	3	6	5
3	2	6	7
4	2	8	9

b 9

c 2

d 2.44

e



f 4

Cumulative frequency ogive

11

Score	Frequency
84	2
85	2
86	4
87	6
88	2

12



13

a 30

b 10

c 10

d 20

e 30

f 20

14

a

i 12

ii 7

iii 17

b

i 675

ii 640

iii 760

c

i 2100

ii 1900

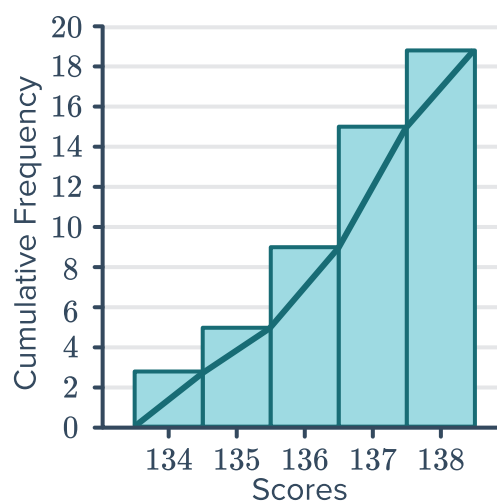
iii 2450

15

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
134	3	3
135	2	5
136	4	9
137	6	15
138	4	19
Total:	19	

b



c 137

d 135



Cumulative frequency and grouped data

16

a

i

Class	Class centre	Frequency	Cumulative frequency
1 – 7	4	11	11
8 – 14	11	14	25
15 – 21	18	10	35
22 – 28	25	15	50
29 – 35	32	24	74
Totals		74	

ii 29 – 35

iii 22 – 28

iv 20.55

b

i

Class	Class centre	Frequency	Cumulative frequency
1 – 9	5	8	8
10 – 18	14	16	24
19 – 27	23	4	28
28 – 36	32	21	49
37 – 45	41	16	65
Totals		65	

ii 28 – 36

iii 28 – 36

iv 25.91

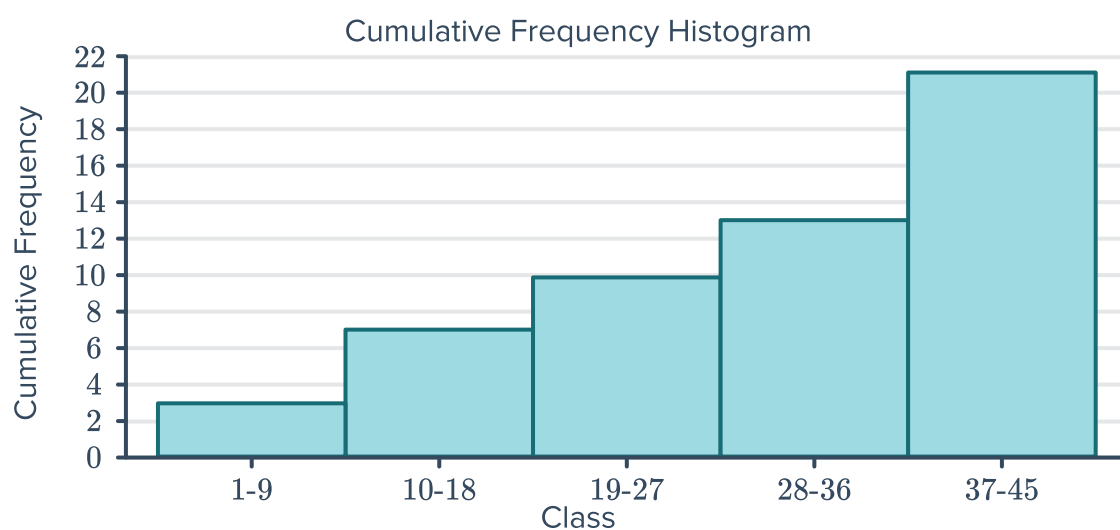
17



a

Class	Class centre	Frequency	Cumulative frequency
1 – 9	5	3	3
10 – 18	14	4	7
19 – 27	23	3	10
28 – 36	32	3	13
37 – 45	41	8	21
Totals		21	

b



c 37 – 45

d 28 – 36

18

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
1 – 4	4	4
5 – 8	7	11
9 – 12	11	22
13 – 16	7	29
17 – 20	4	33

b 33

c 4

d Symmetrical

e 12

19



a

Score (x)	Frequency (f)	Cumulative frequency (cf)
1 – 5	7	7
6 – 10	15	22
11 – 15	8	30
16 – 20	11	41
21 – 25	2	43
26 – 30	1	44

b 44

c 5

d Positively skewed

e 10

20

a

Score (x)	Frequency (f)	Cumulative frequency (cf)
20 – 24	9	9
25 – 29	18	27
30 – 34	37	64
35 – 39	14	78
40 – 44	9	87
45 – 49	6	93
50 – 54	3	96

b 96

c 5

d Positively skewed

e 34

21



a

Class interval	Class centre(cc)	Frequency(f)	$f \times cc$	Cumul. Freq.
51 – 55	53	1	53	1
56 – 60	58	4	232	5
61 – 65	63	2	126	7
66 – 70	68	3	204	10
71 – 75	73	2	146	12
Total		12	761	

b 56 – 60

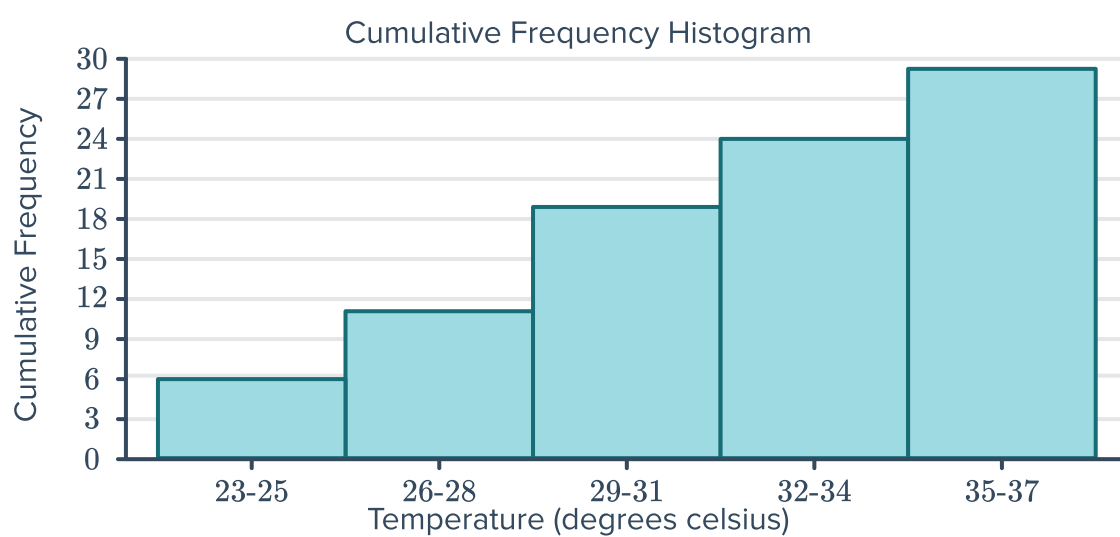
c 63.42

22

a

Class	Class centre	Frequency	Cumulative frequency
23 – 25	24	6	6
26 – 28	27	5	11
29 – 31	30	8	19
32 – 34	33	5	24
35 – 37	36	5	29
Totals:		29	

b



c 29 – 31 °C

d 29.79 °C

23



a

Time Recorded	Frequency	Cumulative frequency
16 : 00 – 16 : 09	4	4
16 : 10 – 16 : 19	2	6
16 : 20 – 16 : 29	6	12
16 : 30 – 16 : 39	7	19
16 : 40 – 16 : 49	5	24
16 : 50 – 16 : 59	1	25

b 25

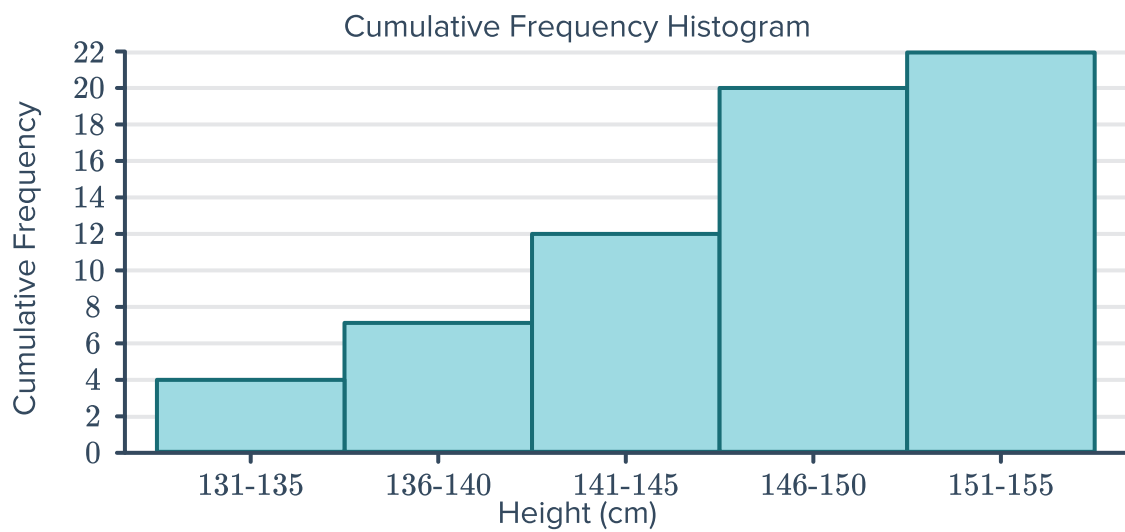
c 16:30 - 16:39

d 24

e 48%

24

a



b 2 students

c 20 students

d 5 students

e 146 – 150

f 148

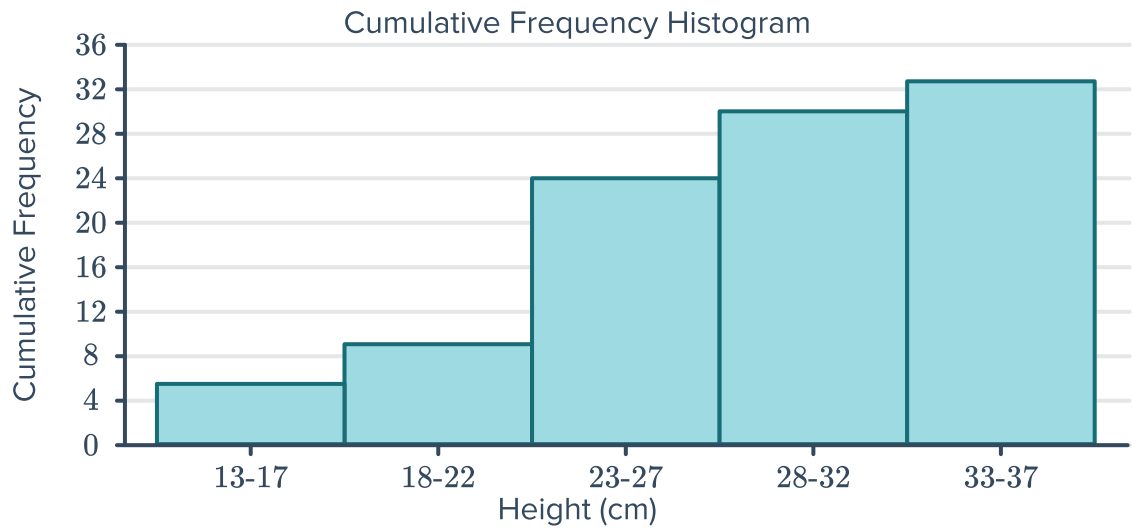
g 141 – 145

25



a 21 cm

b



c 24.55 cm

d 23 – 27

e 23 – 27

f 72.7%

g 27.3%

26

a

i 0.13 seconds

ii 0.132 seconds

b

Reaction Time (seconds)	Class Centre	Frequency	Cumulative frequency
$0.01 \leq t < 0.09$	0.05	61	61
$0.09 \leq t < 0.17$	0.13	70	131
$0.17 \leq t < 0.25$	0.21	69	200

c

i 0.13 seconds

ii 0.133 seconds

d 0.758%